Question	Answer	Marks
13(a)	emission of particles/radiation by unstable nucleus	B1
	spontaneous emission	B1
13(b)(i)	use of graph to determine half-life = 14 minutes	B1
	hence $\lambda = \ln 2 / (14 \times 60)$ (s^{-1})	C1
	N at 14 minutes $= 4.4 \times 10^7$ **and** $A = \lambda N$	C1
	activity $= 4.4 \times 10^7 \times \ln 2 / (14 \times 60)$ $= 3.6 \times 10^4$ Bq	A1
	or	
	correct tangent drawn at time t = 14 minutes	**(B1)**
	magnitude of gradient of tangent identified as activity	**(C1)**
	correct working for gradient leading to activity	**(C1)**
	activity $= 3.6 \times 10^4$ Bq	**(A1)**
13(b)(ii)	$3.6 \times 10^4 = \lambda \times 4.4 \times 10^7$ **or** $\lambda = \ln 2 / (14.0 \times 60)$	C1
	$\lambda = 8.2 \times 10^{-4}$ s^{-1}	A1